

Quantum Field Theory Notes

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Chapter 1

Dirac Equation

- **Aim:** To find Relativistic wave equation for particles
- In non-relativistic approach we write
 1. Hamiltonian of free particle $H = p^2/2m$
 2. Promote the observables to operators and get schrodinger equation

$$\begin{aligned} H &\rightarrow i\hbar \frac{\partial}{\partial t} & p &\rightarrow -i\hbar \nabla \\ i\hbar \frac{\partial \psi}{\partial t} &= -\frac{\hbar^2}{2m} \nabla^2 \psi \end{aligned} \quad (1.1)$$

- if we continue with this approach to get relativistic equation we start with

$$H = \sqrt{p^2 c^2 + m^2 c^4} \quad (1.2)$$

and get the equation

$$i\hbar \frac{\partial \psi}{\partial t} = \left(\sqrt{-\hbar^2 c^2 \nabla^2 + m^2 c^4} \right) \psi \quad (1.3)$$

- demerits of this theory
 1. interpretation of the square-root operator
 2. expanding square-root will lead to non-local theory
 3. space and time coordinates are in asymmetrical form
- we can also not take square root and write

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = (-\hbar^2 c^2 \nabla^2 + m^2 c^4) \psi \quad (1.4)$$

we can write this equation also as

$$\left(\square + \left(\frac{mc}{\hbar} \right)^2 \right) \psi = 0 \quad (1.5)$$

where $\square = \partial_\mu \partial^\mu$

- **Big Problem:** also (1.2) allows negative values of Energy, so energy is not positive definite

1.1 Dirac Equation

- obtain relativistically covariant equation
- positive definite probability density
- linear in time and space derivatives
- hence we can write a equation taking somewhat analogy

$$E = pc + mc^2$$

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{\hbar c}{i} \left(\alpha_1 \frac{\partial \psi}{\partial x_1} + \alpha_2 \frac{\partial \psi}{\partial x_2} + \alpha_3 \frac{\partial \psi}{\partial x_3} \right) + \beta mc^2 \psi \quad (1.6)$$

- α not numbers as equation will not be invariant under rotation
- ψ are column matrix and α 's and β are $N \times N$ matrices

$$i\hbar \frac{\partial \psi_\sigma}{\partial t} = \frac{\hbar c}{i} \left(\alpha_1 \frac{\partial}{\partial x_1} + \alpha_2 \frac{\partial}{\partial x_2} + \alpha_3 \frac{\partial}{\partial x_3} \right)_{\sigma\rho} \psi_\rho + mc^2 \beta_{\sigma\rho} \psi_\rho \quad (1.7)$$

- but this should give us back the Dirac equation

$$-\hbar^2 \frac{\partial^2 \psi_\sigma}{\partial t^2} = (-\hbar^2 c^2 \nabla^2 + m^2 c^4) \psi_\sigma$$

•

$$-\hbar^2 \frac{\partial^2 \psi_\sigma}{\partial t^2} = \left\{ \frac{\hbar c}{i} \left(\alpha_1 \frac{\partial}{\partial x_1} + \alpha_2 \frac{\partial}{\partial x_2} + \alpha_3 \frac{\partial}{\partial x_3} \right)_{\sigma s} + \beta_{\sigma s} mc^2 \right\} \times \\ \left\{ \frac{\hbar c}{i} \left(\alpha_1 \frac{\partial}{\partial x_1} + \alpha_2 \frac{\partial}{\partial x_2} + \alpha_3 \frac{\partial}{\partial x_3} \right)_{sr} + \beta_{sr} mc^2 \right\} \psi_r$$

we can write this as

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = -\hbar^2 c^2 \frac{\alpha_i \alpha_j + \alpha_j \alpha_i}{2} \frac{\partial^2 \psi}{\partial x_i \partial x_j} + \frac{\hbar mc^3}{i} (\alpha_i \beta + \beta \alpha_i) \frac{\partial \psi}{\partial x_i} + \beta^2 m^2 c^4 \psi \quad (1.8)$$

To get back dirac equation we need¹

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij} \quad (1.9)$$

$$\{\alpha_i, \beta_i\} = 0 \quad (1.10)$$

$$\alpha_i^2 = \beta^2 = 1 \quad (1.11)$$

¹ $\{A, B\} = AB + BA$

1.1.1 Lorentz Covariance of Dirac Equation

- We choose the matrices

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (1.12)$$

- now we can write Dirac equation as

$$(i\hbar \not{\partial} - mc)\psi = 0 \quad (1.13)$$

where $\not{\partial} = \gamma^\mu \partial_\mu$

- These γ matrices follow

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \mathbb{1} \quad (1.14)$$

- addition of electromagnetic interaction changes the equation to

$$\left(\not{p} - \frac{e\mathcal{A}}{c} - mc \right) \psi = 0 \quad (1.15)$$

Proof of Covariance

- In frame O' the dirac equation has form

$$(i\hbar \tilde{\gamma}^\mu \partial'_\mu - mc) \psi'(x') = 0 \quad (1.16)$$

- We have tilde over the γ matrices as in O' frame, the observer may write these matrices in some different representation but all such representations are related by unitary transformation

$$\gamma'^\mu = U^\dagger \gamma^\mu U \quad (1.17)$$

where U is some unitary matrix.

- Under this we can always redefine $\psi \rightarrow U\psi$ which gives us

$$\begin{aligned} (i\hbar \tilde{\gamma}^\mu \partial'_\mu - mc) U \psi'(x') &= 0 \\ \text{(Multiply from left by } U^\dagger) & \\ (i\hbar U^\dagger \tilde{\gamma}^\mu U \partial'_\mu - mc) \psi'(x') &= 0 \\ (i\hbar \gamma^\mu \partial'_\mu - mc) \psi'(x') &= 0 \end{aligned}$$

- Hence we can drop distinction between γ and $\tilde{\gamma}$

- We introduce more matrices to prove (1.17)

$$\begin{aligned}
& \mathbb{1} \\
& \gamma^\mu = \gamma^0, \gamma^1, \gamma^2, \gamma^3 \\
& \gamma^{\mu\nu} = \frac{1}{2}[\gamma^\mu, \gamma^\nu] = \gamma^{[\mu}\gamma^{\nu]} : \gamma^{01}, \gamma^{02}, \gamma^{03}, \gamma^{12}, \gamma^{13}, \gamma^{23} \\
& \gamma^{\mu\nu\rho} = \gamma^{[\mu}\gamma^\nu\gamma^{\rho]} = \gamma^{012}, \gamma^{013}, \gamma^{023}, \gamma^{123} \\
& \gamma^{\mu\nu\rho\sigma} = \gamma^{[\mu}\gamma^\nu\gamma^\rho\gamma^{\sigma]} = \gamma^{0123}
\end{aligned}$$

- We start with some properties

1. $\gamma_\mu\gamma_\nu\gamma_\mu = \gamma_\nu$
2. $\text{Tr}(\gamma_\mu) = 0$
3. The all γ_A matrices are linearly independent. (A index for all of the above 4 matrices)

$$\sum_A a_A \gamma^A = 0 \quad (1.18)$$

when all $a_A = 0$

Proof: Suppose they are not linearly independent and one of a_B is zero and others are not such that

$$\begin{aligned}
a_B \gamma^B + \sum_{A \neq B} a_A \gamma^A &= 0 \\
a_B + \sum_{A \neq B} a_A \gamma^A \gamma_B &= 0
\end{aligned}$$

With help of anti-commutation relations (1.14) the product $\gamma_A \gamma_B$ can be written as $b \gamma_C$ with some complex-scalar b

For eg. $\gamma_1 \gamma_2 = \frac{1}{2}[\gamma_1, \gamma_2] = \gamma_{12}$

$$a_B + \sum_C b_C \gamma_C = 0$$

4. Any arbitrary matrix X can be written as a linear combination

$$X = \sum_A a_A \gamma_A \quad (1.19)$$

and

$$a_A = \frac{1}{4} \quad (1.20)$$

- since both Dirac equation and the Lorentz transformation are linear, we demand the transformation between ψ and ψ' to be linear

$$\psi'(x') = \psi'(\Lambda x) = \Lambda_{\frac{1}{2}} \psi(x) = \Lambda_{\frac{1}{2}} \psi(\Lambda^{-1} x') \quad (1.21)$$

- we can write this also as

$$\psi(x) = \Lambda_{\frac{1}{2}}^{-1} \psi'(x') = \Lambda_{\frac{1}{2}}^{-1} \psi'(\Lambda x) \quad (1.22)$$

- Now we manipulate the Dirac equation in the O' frame

$$\begin{aligned} [i\hbar\gamma^\mu\partial'_\mu - mc] \psi'(x') &= 0 \\ [i\hbar\gamma^\mu\Lambda_\mu{}^\nu\partial_\nu - mc] \Lambda_{\frac{1}{2}}\psi(x) &= 0 \\ [i\hbar\Lambda_{\frac{1}{2}}^{-1}\gamma^\mu\Lambda_{\frac{1}{2}}\Lambda_\mu{}^\nu\partial_\nu - mc] \psi(x) &= 0 \end{aligned}$$

- Now for Covariance, we should find $\Lambda_{\frac{1}{2}}$ such that

$$\Lambda_{\frac{1}{2}}^{-1}\gamma^\mu\Lambda_{\frac{1}{2}}\Lambda_\mu{}^\nu = \gamma^\nu$$

or writing it differently

$$\Lambda_{\frac{1}{2}}^{-1}\gamma^\mu\Lambda_{\frac{1}{2}} = \Lambda^\mu{}_\nu\gamma^\nu \quad (1.23)$$

- (1.23) is a fundamental relation for determining $\Lambda_{\frac{1}{2}}$

1.2 Space Reflections

- Study an improper Lorentz Transformation

$$\mathbf{x}' = -\mathbf{x} \quad t' = t \quad (1.24)$$

- The transformation matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = g^{\mu\nu} \quad (1.25)$$

Chapter 2

The Dirac Field

2.1 Quantization of Dirac Field

- Dirac Lagrangian is

$$\mathcal{L} = \bar{\psi} (i\rlap{\not{D}} - m) \psi \tag{2.1}$$

- Hamiltonian is

Chapter 3

Interacting Fields and Feynman Diagrams

3.1 Natural Units

In Natural Units, we set the quantities

$$\hbar = 1 \quad c = 1 \quad k_B = 1 \quad (3.1)$$

But this is just an ansatz, what we actually mean by this is to use units expressed by $\hbar, c$ and k_B . Let us look at some examples

- Consider velocity $v = \frac{c}{2}$ we here drop the c and simply write $v = \frac{1}{2}$
- Entropy: $S = k_B \ln 2^N \rightarrow S = \ln 2^N$
- Action: $S = \hbar \omega t \rightarrow S = \omega t$

Now notice that first case tell us

$$[v] = [1] \iff \left[\frac{L}{T} \right] = [1] \iff [L] = [T] \quad (3.2)$$

that is since velocity is dimensionless the dimension of length and time are same. Similarly, using other cases we imply that

$$[\hbar \omega t] = [Et] \iff [E] = [T]^{-1} = [T]^{-1} \quad (3.3)$$

Also, We have

$$E^2 = p^2 c^2 + m^2 c^4 \rightarrow E^2 = p^2 + m^2$$

This tells us that

$$[E] = [p] = [m] \quad (3.4)$$

$$E = \frac{3}{2} k_B T \rightarrow E = \frac{3}{2} T$$

$$[E] = [T] \quad (3.5)$$

3.1.1 Mass Dimensions

In Quantum Field Theory it is customary to have only one fundamental unit that of mass, all others are defined by using (3.1).

Hence we can have a nice table: The construction of table would be clear from

Quantity	Dimension	Conversion factor
Mass	$[M]$	$1/c^2$
Length	$[M]^{-1}$	$\hbar c$
Time	$[M]^{-1}$	$\hbar$
Energy	$[M]$	1
Momentum	$[M]$	$1/c$
Force	$[M]^2$	$1/(\hbar c)$
Action	$[M]^0$	$\hbar$
Electric charge	$[M]^0$	$\sqrt{\hbar c}$

Table 3.1: Dimensions and conversion factors in natural units

the following formulas

- Mass dimension of mass is trivial
- Mass dimension of length can be deduced using (3.2) and (3.3), (3.4)
- For Force use definition dp/dt which give $[M] / [M]^{-1} = [M]^2$
- Action has units of Energy x Time which is dimensionless in mass dimension
- For Electric charge use coulombs law in Gaussian units $F = \frac{q_1 q_2}{r^2}$

3.2 Perturbation Theory

Let us see Lagrangian of some interacting field theories

- Our first theory is ϕ^4 theory, and the lagrangian is given by

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 \quad (3.6)$$

Where λ is dimensionless coupling constant.

- The equation of motion for ϕ^4 theory will be

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} &= 0 \\ \partial_\mu \partial^\mu \phi + m^2 \phi + \frac{\lambda}{3!} \phi^3 &= 0 \end{aligned} \quad (3.7)$$

- Quantum Electrodynamics:

$$\begin{aligned}\mathcal{L}_{\text{QED}} &= \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{Maxwell}} + \mathcal{L}_{\text{int}} \\ &= \bar{\psi}(i\not{\partial} - m)\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} - e\bar{\psi}\gamma^\mu\psi A_\mu\end{aligned}\quad (3.8)$$

3.3 Perturbation Expansion of Correlation Functions

We now start the study of correlation function in the interaction fields, We will calculate the two point correlation function¹

$$\langle\Omega|T\phi(x)\phi(y)|\Omega\rangle\quad (3.9)$$

- Physically it denotes the propagation amplitude of particle between y and x
- We use $|\Omega\rangle$ as the ground state of interaction theory need not to be $|0\rangle$
- In free theory it is simply Feynman propagator:

$$\langle 0|T\phi(x)\phi(y)|0\rangle_{\text{free}} = \int \frac{d^4p}{(2\pi)^4} \frac{ie^{ip\cdot(x-y)}}{p^2 - m^2 + i\epsilon}\quad (3.10)$$

3.4 Wick's Theorem

- Gives an Easy way to evaluate $\langle 0|T\{\phi_I(x_1)\phi_I(x_2)\cdots\phi_I(x_n)|0\rangle$
- First decompose the field in positive and negative frequency parts as

$$\phi_I(x) = \phi_I^+(x) + \phi_I^-(x)\quad (3.11)$$

$$\phi_I^+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} a_{\mathbf{p}} e^{-ip\cdot x} \quad \phi_I^-(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} a_{\mathbf{p}}^\dagger e^{ip\cdot x}\quad (3.12)$$

This tells us that

$$\phi_I^+(x)|0\rangle = 0 \quad \text{and} \quad \langle 0|\phi_I^-(x)\quad (3.13)$$

Which simplifies a lot of work for us

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¹Recall from Quantum Mechanics that correlation function is defined by $C(t) = \langle x(t)x(0) \rangle$

Wicks Theorem

$$T\{\phi_I(x_1)\phi_I(x_2)\cdots\phi_I(x_n)\} = N\{\phi_I(x_1)\phi_I(x_2)\cdots\phi_I(x_n) + \text{all possible contractions}\}$$

Proof:

- Use Mathematical Induction
- We have already checked the case with $n = 2$
- Assume theorem is true for $n - 1$ fields. Also we choose $x_1^0 \geq x_2^0 \cdots \geq x_n^0$

Appendix A: Greens Functions

Greens functions are extensively used in the Quantum Field theory. Let's review them in detail here.

3.5 Introduction

Suppose we have a differential equation

$$\mathcal{D}_x f(x) = g(x) \quad (3.14)$$

where $\mathcal{D}_x$ is a differential operator. To formally solve the problem just invert the operator

$$\begin{aligned} D |f\rangle &= |g\rangle \\ \implies |f\rangle &= D^{-1} |g\rangle + \sum c_i |h_i\rangle \end{aligned} \quad (3.15)$$

Where the two terms are like particular integral and complementary functions respectively and $D |h_i\rangle = 0 \quad \forall i$.

Clearly, D^{-1} will be some integral operator and we can find $f(x)$ as follows

$$\begin{aligned} f(x) &= \langle x | f \rangle = \langle x | D^{-1} | g \rangle + \sum c_i \langle x | h_i \rangle \\ f(x) &= \int dx' \langle x | D^{-1} | x' \rangle \langle x' | g \rangle + \sum c_i \langle x | h_i \rangle \\ f(x) &= \int dx' G(x, x') g(x') + \sum c_i h_i(x) \end{aligned} \quad (3.16)$$

where $G(x, x') = \langle x | D^{-1} | x' \rangle$ which is our *Greens Function*.

$$DD^{-1} = \mathbb{1}$$

in position space this would be

$$\langle x | DD^{-1} | x' \rangle = \langle x | x' \rangle$$

$$\begin{aligned}
\int dy \langle x | D | y \rangle \langle y | D^{-1} | x' \rangle &= \langle x | x' \rangle \\
\int dy \delta(x - y) \mathcal{D}_y G(x, y) &= \langle x | x' \rangle \\
\mathcal{D}_x G(x, x') &= \delta(x - x')
\end{aligned} \tag{3.17}$$

3.6 Greens functions in 1D

Suppose we want to solve the Linear Ordinary Differential Equation

$$y'' + p(x)y' + q(x)y = f(x) \tag{3.18}$$

for $a < x < b$. To solve this using Greens function we follow the following procedure:

1. First consider $G(x, x'')'' + p(x)'G(x, x') + q(x)G(x, x') = \delta(x - x')$. Where $G(x, x')$ is Greens Function.
2. Find the solution of $y'' + p(x)y' + q(x)y = 0$. Suppose $p(x)$ and $q(x)$ are constants, then we can write $y(x) = Ay_1(x) + By_2(x)$. Write using boundary conditions

$$G(x, x') = \begin{cases} Ay_1(x')y_2(x) & \text{if } a < x < x' \\ Ay_2(x')y_1(x) & \text{if } x' < x < b \end{cases} \tag{3.19}$$

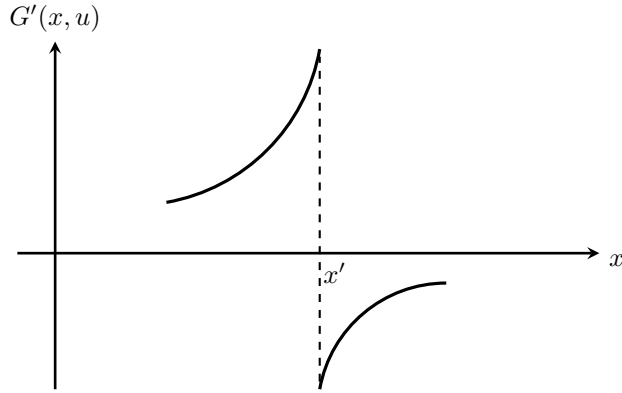
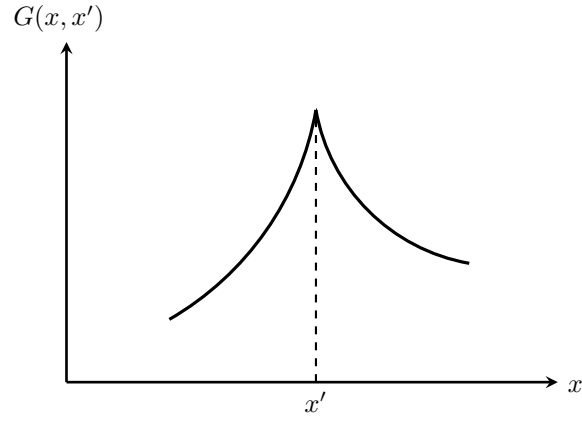
3. Find A using

$$\begin{aligned}
\lim_{\epsilon \rightarrow 0} \left[\left[\frac{dG}{dx} \right]_{x'-\epsilon}^{x'+\epsilon} + \int_{x'-\epsilon}^{x'+\epsilon} p(x) \frac{dG}{dx} dx + \int_{x'-\epsilon}^{x'+\epsilon} q(x) G dx \right] = \\
\int_{x'-\epsilon}^{x'+\epsilon} \delta(x - x') dx \tag{3.20}
\end{aligned}$$

4. Then

$$y(x) = Ay_2(x) \int_a^x y_1(x') f(x') dx' + Ay_1(x) \int_x^b y_2(x') f(x') dx' \tag{3.21}$$

Note here that the second derivative of Greens function is Dirac Delta and hence the first derivative should have a finite jump discontinuity at x' . And hence Greens function must be having a cusp at x' .



Let's consider an example

$$-y''(x) = f(x) \quad y(0) = y(1) = 0 \quad (3.22)$$

Solving the homogeneous differential equation we get solution

$$y = c_0 + c_1 x \quad (3.23)$$

$$G(x, x') = \begin{cases} Ax(1 - x') & 0 < x < x' \\ A(1 - x)x' & x' < x < 1 \end{cases} \quad (3.24)$$

Now using step 3

$$\begin{aligned} A[x' - (1 - x')] &= -1 \\ A &= 1 \end{aligned} \quad (3.25)$$

Therefore,

$$G(x, x') = \begin{cases} x(1 - x') & 0 < x < x' \\ (1 - x)x' & x' < x < 1 \end{cases} \quad (3.26)$$

Now if $f(x) = \sin x$ Then

$$y(x) = x \int_0^x (1-x') \sin \pi x' dx' + (1-x) \int_x^1 x' \sin \pi x' dx' \quad (3.27)$$

] =

3.7 Green's Theorem

It is just simplification of Divergence Theorem but useful.

$$\oint_S \mathbf{A} \cdot \mathbf{n} da = \int_V \nabla \cdot \mathbf{A} d^3x \quad (3.28)$$

Take the field $\mathbf{A} = \phi \nabla \psi$. Hence

$$\nabla \cdot (\phi \nabla \psi) = \nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi \quad (3.29)$$

and

$$\phi \nabla \psi \cdot \mathbf{n} = \phi \frac{\partial \psi}{\partial n} \quad (3.30)$$

Therefore, the divergence theorem now becomes

$$\oint_S \phi \frac{\partial \psi}{\partial n} da = \int_V (\nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi) d^3x \quad (3.31)$$

Interchange ϕ and ψ we get

$$\oint_S \psi \frac{\partial \phi}{\partial n} da = \int_V (\nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi) d^3x \quad (3.32)$$

Subtracting two we get *Green's Theorem*

$$\oint_S \left(\psi \frac{\partial \phi}{\partial n} - \phi \frac{\partial \psi}{\partial n} \right) da = \int_V (\psi \nabla^2 \phi - \phi \nabla^2 \psi) d^3x \quad (3.33)$$

We can convert Poisson's Differential Equation to integral equation using Greens Theorem

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0} \quad (3.34)$$

Choose $\phi = \Phi$ and $\psi = 1/|\mathbf{x} - \mathbf{x}'| = 1/R$ and put in (3.33)

$$\oint_S \left(\frac{1}{R} \frac{\partial}{\partial n'} \Phi(\mathbf{x}') - \Phi(\mathbf{x}') \frac{\partial}{\partial n'} \frac{1}{R} \right) da = \int_V \left(-\frac{1}{R} \frac{\rho(\mathbf{x}')}{\epsilon_0} + 4\pi \delta^3(\mathbf{x} - \mathbf{x}') \Phi(\mathbf{x}') \right) d^3x$$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{x}')}{R} d^3x + \frac{1}{4\pi} \oint_S \left(\frac{1}{R} \frac{\partial}{\partial n'} \Phi(\mathbf{x}') - \Phi(\mathbf{x}') \frac{\partial}{\partial n'} \frac{1}{R} \right) da \quad (3.35)$$

3.8 Green's Function in 3D: Poisson Equation

Suppose we want to solve the Poisson equation

$$\nabla^2 \Phi(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon_0}$$

Greens function in this case would satisfy

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = -4\pi\delta^{(3)}(\mathbf{x} - \mathbf{x}') \quad (3.36)$$

Which we easily identify to be

$$G(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|} + F(\mathbf{x}, \mathbf{x}') \quad (3.37)$$

Where $F(\mathbf{x}, \mathbf{x}')$ satisfy

$$\nabla^2 F(\mathbf{x}, \mathbf{x}') = 0 \quad (3.38)$$

Now in (3.35) put $\phi = \Phi$ and $\psi = G(\mathbf{x}, \mathbf{x}')$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G(\mathbf{x}, \mathbf{x}') d^3x' + \frac{1}{4\pi} \oint_S \left(G(\mathbf{x}, \mathbf{x}') \frac{\partial}{\partial n'} \Phi(\mathbf{x}') - \Phi(\mathbf{x}') \frac{\partial}{\partial n'} G(\mathbf{x}, \mathbf{x}') \right) da \quad (3.39)$$

Now silimilar to 1D case of differential equations where we require the value of function or its derivatives to determine the solution uniquely, We have boundary conditions in 3D case too Boundary Conditions:

- *Dirichlet Boundary Condition*

$$G_D(\mathbf{x}, \mathbf{x}') = 0 \quad \text{for } \mathbf{x}' \text{ on } S \quad (3.40)$$

- *Neumann Boundary Condition*

$$\frac{\partial}{\partial n} G_N(\mathbf{x}, \mathbf{x}') = -\frac{4\pi}{S} \quad \text{for } \mathbf{x}' \text{ on } S \quad (3.41)$$

3.9 Green's Function for wave Equation

- We start with generalized wave equation

$$\nabla^2 \psi - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = -4\pi f(\mathbf{x}, t) \quad (3.42)$$

- We remove the explicit time dependence using Fourier transforms of ψ and f

$$\psi(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(\mathbf{x}, \omega) e^{-i\omega t} d\omega \quad (3.43)$$

$$f(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\mathbf{x}, \omega) e^{-i\omega t} d\omega \quad (3.44)$$

- Now putting this in (3.42), we get

$$(\nabla^2 + k^2)\psi(\mathbf{x}, \omega) = -4\pi f(\mathbf{x}, \omega) \quad (3.45)$$

with $k = \omega/c$.

- Now the Green's function appropriate to our problem will satisfy

$$(\nabla^2 + k^2)G_k(\mathbf{x}, \mathbf{x}') = -4\pi\delta(\mathbf{x}, \mathbf{x}') \quad (3.46)$$

- Assume there are no boundary surfaces $\implies$ Spherically symmetric Green's function,

$$\begin{aligned} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dG_k}{dr} \right) + k^2 G_k &= -4\pi\delta(\mathbf{r}) \\ \frac{2}{r} \frac{dG_k}{dr} + \frac{d^2 G_k}{dr^2} + k^2 G_k &= -4\pi\delta(\mathbf{r}) \\ \frac{1}{r} \frac{d^2}{dr^2} (rG_k) + k^2 G_k &= -4\pi\delta(\mathbf{r}) \end{aligned}$$

- for $\mathbf{r} \neq 0$

$$\frac{1}{r} \frac{d^2}{dr^2} (rG_k) + k^2 (rG_k) = 0 \quad (3.47)$$

with solution

$$rG_k = Ae^{ikr} + Be^{-ikr} \quad (3.48)$$

- When $r \rightarrow 0$ we take help from electrostatics and compare situation with point charge at origin whose Green's function is known to be $1/r$. Hence,

$$\lim_{\mathbf{r} \rightarrow 0} G_k(\mathbf{r}) = \frac{1}{r} \quad (3.49)$$

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$$G_k(\mathbf{r}) = AG_k^+(\mathbf{r}) + BG_k^-(\mathbf{r}) \quad (3.50)$$

with

$$G_k^\pm(\mathbf{r}) = \frac{e^{\pm ikr}}{r} \quad (3.51)$$

- $A+B=1$. The G_k^+ represents diverging outgoing wave and G_k^- represents converging incoming wave.